Maths

Probability

1. Probabilistic graphical models can only do 2 things:
   1. Inference: calculating a probability of some data modelled as RV using the model $\mathcal{M}$:
      1. evaluation $P(obs|state)$,
      2. decoding $P(state|obs)$
   2. Learning model parameters: refers to finding the graphical model $\mathcal{M}$, given some data $\mathcal{D}$.
      1. MAP (maximum a posteriori)
      2. Maximum likelihood
      3. expectation maximization
      4. forward-backward algorithm
2. Conditional probability can be computed using:
   1. Bayes rule
   2. Enumeration
3. θ is model parameter.
4. For example RV x is binomial, x is the number of times a coin landed heads in N trials.
   1. $$P(x = X|\theta = p) = \binom{N}{x} p^x (1-p)^{N-x} \quad \text{(likelihood)}$$
5. if RV x is single variable Gaussian, for modeling a real value, e.g. temperature sensor reading.
   1. $$P(x = X|\theta = (\mu, \sigma^2)) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad \text{(likelihood)}$$
6. Note: Notice that in both cases the RV x ended up in the exponent.
7. $$p(\theta|x = D) = \frac{p(x|\theta)p(\theta)}{p(x)}$$

   $p(x|\theta)$ is the likelihood, $p(\theta)$ is the prior distribution, $p(\theta|x = D)$ is the posterior, and $p(x)$ is the normalization term.

Probabilistic Inference Methods

1. Maximum Likelihood Estimation (MLE)
   1. $x^* = \arg\max p(d|x)$.
   2. Likelihood: $p(d|x)$
   3. $x*$ is the most probable value of x.
   4. $\{x\}$ is a set of possible values of x (the candidates of x).
   5. $\arg\max$ to return the value of x that maximizes $p(d|x)$.
   6. The weakness of MLE is that it allows impossible candidates, as it treats candidates uniformly.
   7. The intuitive meaning of MLE:
      1. Take each possible illness e.g. x=flu.
      2. Verify if it fits to the symptoms $\rightarrow$ probability of x.
      3. Decide the illness based on the hights probability.
2. Maximum A Posteriori (MAP)
   1. $$\begin{aligned} x^* &= \arg\max p(x|d) \\ &= \arg\max p(d|x)p(x) \end{aligned}$$

      . Likelihood:
   2. MAP treats the candidates differently: $p(x = \text{flu}) > p(x = \text{cancer})$
   3. Main criticism of MAP is how to determine $p(x)$.
3. Full Bayesian Inference
   1. $p(x|d) = \frac{p(d|x)p(x)}{p(d)}$. $P(d)$ is intractable to solve.
   2. This is the ideal and most robust inference that besides providing a prediction on the value of x, it also provides the uncertainty, or how high/low the probability of the prediction.

   3. It's possible because of $p(d)$, the evidence. This will normalize the product of $p(d|x)p(x)$.
   4. However to obtain $p(d)$ is analytically intractable, and in discrete, it can be prohibitively expensive.
4. Example: According to our observation, a patient has a set of symptoms: d : {stiff neck, coughing, headache, fever}. We also have a set of possible illness: flu, meningitis, infection and cancer.

Lagrange Multiplier [3]

Hungarian Matching [2]

References:

1. Harrison H. Barrett, Kyle J. Myers. Foundations of Image Science.
2. Discrete Mathematics, John Dossey, 1987.
3. Machine Learning Fundamentals: A Concise Introduction, Hui Jiang.
4. Neural Message Passing
5. Lecture 7, Stereo + MRF, EE5731 Visual Computing